

# Chapter 24: Electromagnetic Waves

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PHY201  
Lecture 24



## Part I

# Maxwell's Equations: Prediction of EM Waves

# Outline of Part I: Maxwell's Equations: Prediction of EM Waves

A Historical Puzzle

Maxwell's Four Equations

Speed of Electromagnetic Waves

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# Light as a Particle, as a Wave, and as a Puzzle

How can light travel through empty space?

**Sound waves** are described by oscillatory motion in **air**.

**Water waves** are described by oscillatory motion in **water**.

Experiments reveal that **light behaves as a wave**.

But light reaches us from the Sun through **empty space**; no matter is present.

So if **nothing** is there, **what** is in oscillatory motion?

## Wave Types Compared

- ▶ **Sound:** oscillating pressure in matter.
- ▶ **Water waves:**  
oscillating surface of a fluid.
- ▶ **Electromagnetic (EM) waves:**  
oscillating *electric and magnetic fields*.



EM waves do *not* require matter.

The **fields** carry energy through space, not matter.

This was Maxwell's revolutionary insight.

## Before Maxwell: Two Separate Sciences

### The Historical Situation in the 1850s

- ▶ Electricity and magnetism were treated as completely separate subjects
- ▶ Coulomb, Ampère, and Faraday had each developed their own laws
- ▶ No one recognized electricity and magnetism as aspects of a single force
- ▶ Faraday had shown: changing  $\vec{B}$  induces  $\vec{E}$
- ▶ Ampère's law: electric currents create  $\vec{B}$

### The Asymmetry Maxwell Noticed

- ▶ Faraday: changing  $B \Rightarrow E$  ✓
- ▶ Ampère: current  $\Rightarrow B$  ✓
- ▶ Reverse of Faraday: changing  $E \Rightarrow B$ ?  
**missing!**

Maxwell argued that *symmetry demands* a changing electric field must also produce a magnetic field — completing the picture.

# Outline of Part I: Maxwell's Equations: Prediction of EM Waves

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# Maxwell's Four Equations (Conceptual)

## The Four Laws that Unify Electromagnetism

1. **Gauss's Law:** Electric charges produce electric fields.
2. **Gauss's Law for Magnetism:** No isolated magnetic monopoles have ever been observed — magnetic field lines always form closed loops.
3. **Faraday's Law:** A *changing* magnetic field induces an electric field.
4. **Ampère–Maxwell Law:** Magnetic fields are produced by (a) electric currents and (b) *changing* electric fields.

## Maxwell's Critical Addition

Clause (b) of the Ampère–Maxwell Law was Maxwell's own contribution — the **displacement current** term. It gives the theory its self-sustaining wave solution.



# Maxwell's Key Insight: The Missing Symmetry

## Faraday + Maxwell = Mutual Feedback

$$\text{Faraday's Law: } \Delta \vec{B} / \Delta t \Rightarrow \vec{E}$$

$$\text{Ampère–Maxwell Law: } \Delta \vec{E} / \Delta t \Rightarrow \vec{B}$$

## Why This Creates a Wave

- ▶ A changing  $\vec{E}$  at one location creates  $\vec{B}$  in nearby space
- ▶ That changing  $\vec{B}$  creates  $\vec{E}$  slightly farther away
- ▶ The disturbance propagates outward continuously
- ▶ **No medium is needed** — only fields

## Crucial Distinction

*Static* fields do **not** regenerate one another.

Only *changing* (time-varying) fields drive the feedback loop.

# What Is Actually Oscillating?

In an electromagnetic wave, the **electric and magnetic field values** oscillate at every point in space and time. No material substance moves along with the wave.

## Sound Wave vs. EM Wave

- ▶ **Sound:** molecules oscillate back and forth in the direction of travel; *requires particles*
- ▶ **EM wave:**  $\vec{E}$  and  $\vec{B}$  values oscillate at each point in space; *requires fields only*

Removing all matter from space leaves the *fields* intact. EM waves travel through perfect vacuum because the fields themselves are the “medium.”

**Important:** EM waves are *transverse* — the field oscillations are *perpendicular* to the direction of travel.

Unlike sound, which is a longitudinal (compression) wave.

# Outline of Part I: Maxwell's Equations: Prediction of EM Waves

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# Speed of Electromagnetic Waves

Maxwell's Prediction — SI Unit:  $\frac{\text{m}}{\text{s}}$

Solving Maxwell's equations produces a wave that travels at speed:

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

where  $\mu_0 = 4\pi \times 10^{-7} \frac{\text{T}\cdot\text{m}}{\text{A}}$  and  $\epsilon_0 = 8.85 \times 10^{-12} \frac{\text{C}^2}{\text{N}\cdot\text{m}^2}$ .

## Variable Definitions

- ▶  $c$ : speed of EM waves in vacuum
- ▶  $\mu_0$ : permeability of free space
- ▶  $\epsilon_0$ : permittivity of free space

## Historical Significance

Maxwell calculated  $c \approx 3.00 \times 10^8 \frac{\text{m}}{\text{s}}$

The measured speed of light was *already known* to be  $\approx 3.00 \times 10^8 \frac{\text{m}}{\text{s}}$ .

**Light is an electromagnetic wave.**

## Example: Speed of Light from Fundamental Constants

### Predicting the Speed of Light

Using  $\mu_0 = 4\pi \times 10^{-7} \frac{\text{T}\cdot\text{m}}{\text{A}}$  and  $\epsilon_0 = 8.85 \times 10^{-12} \frac{\text{C}^2}{\text{N}\cdot\text{m}^2}$ , calculate the predicted speed of electromagnetic waves.

$$\begin{aligned} c &= \frac{1}{\sqrt{\mu_0 \epsilon_0}} \\ &= \frac{1}{\sqrt{(4\pi \times 10^{-7})(8.85 \times 10^{-12})}} \\ &= \frac{1}{\sqrt{1.113 \times 10^{-17}}} \end{aligned}$$

$$c \approx 3.00 \times 10^8 \frac{\text{m}}{\text{s}}$$

This predicted value closely matched the measured speed of light — one of the most profound results in the history of physics.

Light, electricity, and magnetism are **one unified phenomenon**.

## Check Your Understanding

What was Maxwell's crucial addition to Ampère's Law — the idea that was *not* already known from Faraday or Ampère?

Come to lecture to see the answer.

## Check Your Understanding

Why was the equation

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \approx 3.00 \times 10^8 \frac{\text{m}}{\text{s}}$$

so important historically?

Come to lecture to see the answer.

## Part II

# Production and Geometry of Electromagnetic Waves



# Outline of Part II: Production and Geometry of Electromagnetic Waves

Accelerating Charges and Radiation

EM Wave Geometry

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Accelerating Charges and Radiation

EM Wave Geometry

# Which Charges Radiate?

## Charge Motion and Radiation

- ▶ A **stationary** charge: creates a static  $\vec{E}$ , no  $\vec{B}$ , *no radiation*
- ▶ A charge at **constant velocity**: creates both  $\vec{E}$  and  $\vec{B}$ , *no radiation*
- ▶ An **accelerating** charge: changing  $\vec{E}$  and  $\vec{B}$  *radiate energy outward*

## Why Acceleration Is Required

- ▶ Only *changing* fields feed the mutual-regeneration loop
- ▶ Constant velocity: fields change shape but don't generate propagating waves
- ▶ Oscillating (accelerating) charges: create continuously changing fields  $\Rightarrow$  sustained radiation

## Common Misconception

A stationary charge does *not* produce radiation even though it has an electric field.

A charge moving at constant speed does *not* radiate.

Radiation requires *acceleration*.

# Oscillating Dipole Antenna

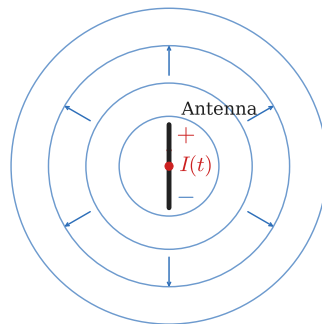
## How a Radio Antenna Works

- ▶ AC current drives electrons up and down:  
oscillating charge = accelerating charge  
 $\Rightarrow$  radiation
- ▶ EM waves emitted at the driving frequency  $f$
- ▶ Waves propagate outward carrying energy

Emitted wavelength:  $\lambda = c/f$ .

Half-wave antenna length:  $\ell = \lambda/2$ .

*EM waves radiate outward*



# Outline of Part II: Production and Geometry of Electromagnetic Waves

Accelerating Charges and Radiation

EM Wave Geometry

# Geometry of a Plane Electromagnetic Wave

## Three Fundamental Geometric Facts

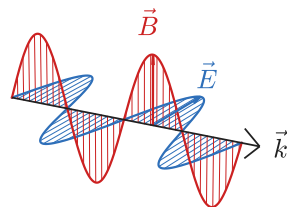
1.  $\vec{E} \perp \vec{B}$  (the two fields are perpendicular to each other)
2. Both  $\vec{E}$  and  $\vec{B}$  are perpendicular to the direction of propagation
3. The fields oscillate *in phase* (peaks and zeros occur together)

## Propagation Direction

The wave travels in the direction of:

$$\vec{E} \times \vec{B}$$

This cross-product points *away* from the source, carrying energy outward.



$$\vec{k} = \vec{E} \times \vec{B}$$

## Electric and Magnetic Field Ratio

### Field Amplitude Relationship — SI Unit: $\frac{\text{V}}{\text{m}} / \text{T}$

In a plane electromagnetic wave, the electric and magnetic field amplitudes satisfy:

$$c = \frac{E}{B}$$

where  $E$  and  $B$  are the instantaneous (or peak) field magnitudes.

### Physical Interpretation

- ▶  $E$  is numerically much larger than  $B$  (since  $c \sim 3 \times 10^8 \frac{\text{m}}{\text{s}}$ ).
- ▶ Despite this, both fields carry *equal* amounts of energy.
- ▶ The ratio is fixed by the wave speed — it is not a free parameter.

From  $c = E/B$ :

$$B = \frac{E}{c}, \quad E = cB$$

Either field completely determines the other.

# The EM Wave Equation

## Frequency, Wavelength, and Wave Speed

All electromagnetic waves in vacuum obey:

$$c = f\lambda$$

where  $c = 3.00 \times 10^8 \frac{\text{m}}{\text{s}}$  is the **same** for every EM wave in vacuum.

### Key Insight

- ▶ All EM waves travel at *the same speed* in vacuum
- ▶ Frequency and wavelength adjust *inversely*
- ▶ High frequency  $\Rightarrow$  short wavelength
- ▶ Low frequency  $\Rightarrow$  long wavelength

### 💀 Common Misconception

Higher-frequency waves do **not** travel faster.

Radio waves and gamma rays both travel at  $c$  in vacuum.

Frequency determines energy per photon (quantum), not speed.



## Example: Finding the Magnetic Field Amplitude

### EM Wave Field Amplitudes

An EM wave has a peak electric field  $E_0 = 1000 \frac{\text{V}}{\text{m}}$ . Find (a) the peak magnetic field  $B_0$ , and (b) the wavelength if the frequency is  $f = 100 \text{ MHz}$ .

$$\begin{aligned} \text{(a)} \quad B_0 &= \frac{E_0}{c} \\ &= \frac{1000}{3.00 \times 10^8} \end{aligned}$$

$$B_0 = 3.33 \times 10^{-6} \text{ T} = 3.33 \mu\text{T}$$

$$\begin{aligned} \text{(b)} \quad \lambda &= \frac{c}{f} \\ &= \frac{3.00 \times 10^8}{100 \times 10^6} \end{aligned}$$

$$\lambda = 3.00 \text{ m}$$

$B_0$  is tiny ( $\sim \mu\text{T}$ ) yet both  $E$  and  $B$  fields carry equal energy.

$\lambda = 3 \text{ m}$  is in the FM radio band.

$B_0$  is small numerically because  $c$  is very large.

## Check Your Understanding

An EM wave travels in the  $+z$  direction. The electric field points in the  $+x$  direction.

In which direction does the magnetic field point?

Come to lecture to see the answer.

## Check Your Understanding

Why can electromagnetic waves travel through a perfect vacuum, when sound waves cannot?

Come to lecture to see the answer.

## Check Your Understanding

Do radio waves (frequency  $\sim 1$  MHz) travel slower than gamma rays (frequency  $\sim 10^{20}$  Hz) in vacuum?

Come to lecture to see the answer.

## Part III

# The Electromagnetic Spectrum

# Outline of Part III: The Electromagnetic Spectrum

The Spectrum as a Continuum

Radio Waves and Microwaves

Infrared, Visible, and Ultraviolet

X-Rays and Gamma Rays

# Outline of Part III: The Electromagnetic Spectrum

The Spectrum as a Continuum

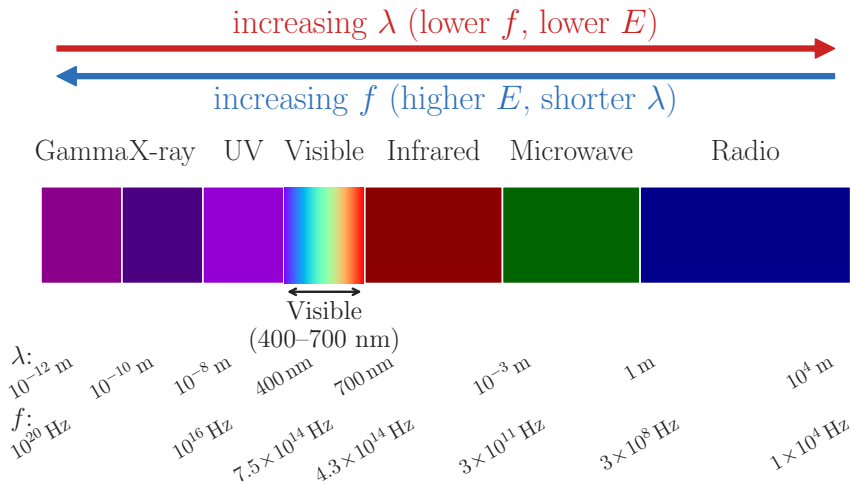
Radio Waves and Microwaves

Infrared, Visible, and Ultraviolet

X-Rays and Gamma Rays

# The Electromagnetic Spectrum

All EM waves obey  $c = f\lambda$  with the same  $c$ . They differ only in frequency (and therefore wavelength).





# Three Unifying Trends Across the Spectrum

## Frequency Determines Behavior

As frequency **increases** (wavelength decreases):

### Energy per Photon

- ▶ Higher frequency  $\Rightarrow$  higher photon energy
- ▶ Can ionize atoms and break chemical bonds
- ▶ Explains biological hazards of UV, X-rays, gamma rays

### Spatial Resolution

- ▶ Shorter wavelength  $\Rightarrow$  can resolve smaller structures
- ▶ X-rays image bone and soft tissue details
- ▶ Radio waves cannot resolve small objects

### Information Bandwidth

- ▶ Higher frequency carriers  $\Rightarrow$  more information per second
- ▶ Fiber optics (IR/visible) outperform radio
- ▶ 5G uses higher frequencies than 4G for this reason

# Outline of Part III: The Electromagnetic Spectrum

The Spectrum as a Continuum

Radio Waves and Microwaves

Infrared, Visible, and Ultraviolet

X-Rays and Gamma Rays

# Radio Waves

## Properties

- ▶ Longest wavelengths in the EM spectrum
- ▶ Produced by oscillating currents in antennas
- ▶ Penetrate buildings and the lower atmosphere effectively
- ▶ Frequency range:  $\sim 3 \text{ kHz}$  to  $300 \text{ MHz}$

## Applications

- ▶ AM/FM radio broadcasting
- ▶ Television transmission
- ▶ WiFi, Bluetooth, GPS
- ▶ MRI (uses radio-frequency pulses)
- ▶ Radar (overlaps with microwave)

## Why radio for MRI?

MRI uses radio-frequency EM pulses to flip nuclear spins in hydrogen atoms. The body's own protons emit radio waves as they return to equilibrium.

No ionizing radiation is involved.

# Microwaves

## Properties

- ▶ Wavelengths from  $\sim 1$  mm to  $\sim 30$  cm
- ▶ Produced by magnetrons and klystrons
- ▶ Can penetrate clouds (useful for radar)

## Applications

- ▶ **Microwave ovens:** 2.45 GHz excites water-molecule rotations, generating heat
- ▶ Radar: aircraft, weather, speed guns
- ▶ Satellite and cell phone communication
- ▶ Cosmic Microwave Background (CMB) astronomy

## Microwave-Oven mechanism:

Water molecules are polar — they rotate to align with the rapidly oscillating  $\vec{E}$  field. This rotational motion is heat.

Dry foods with little water heat slowly.

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X-Rays and Gamma Rays

# Infrared Radiation

## Properties

- ▶ Wavelengths:  $\sim 700$  nm to  $\sim 1$  mm
- ▶ Emitted by any object with temperature above 0 K
- ▶ Associated with molecular vibrations and thermal radiation
- ▶ Human body radiates primarily in the infrared

## Applications

- ▶ Thermal imaging (medical, military)
- ▶ Remote controls; night vision
- ▶ Fiber optic communication (1550 nm)

The **greenhouse effect** involves IR.

Sunlight warms Earth's surface, which re-radiates as IR.

Greenhouse gases absorb this IR, trapping heat.

# Visible Light

## Properties

- ▶ Wavelengths:  $\sim 400$  nm (violet) to  $\sim 700$  nm (red)
- ▶ Only a narrow slice of the entire EM spectrum
- ▶ Produced by atomic electron transitions

## Color Order: ROYGBIV

- ▶ **Red**: longest  $\lambda$ , lowest frequency
- ▶ **Violet**: shortest  $\lambda$ , highest frequency
- ▶ Higher frequency  $\Rightarrow$  higher photon energy

## Visible Light: Not Special

Visible light is *not* fundamentally different from radio waves or X-rays.

It is simply the narrow frequency range detectable by the human eye.

**The same physical laws govern all EM radiation.**

# Ultraviolet Radiation

## Properties

- ▶ Wavelengths:  $\sim 10$  nm to  $\sim 400$  nm
- ▶ Higher photon energy than visible light
- ▶ Can ionize molecules and break chemical bonds
- ▶ Classified: UV-A (400–315 nm), UV-B (315–280 nm), UV-C (<280 nm)

## Beneficial Effects

- ▶ UV-B triggers vitamin D synthesis in skin
- ▶ UV-C kills bacteria (sterilization)
- ▶ Fluorescence and black-light applications

## Hazards

- ▶ Damages DNA  $\Rightarrow$  sunburn, cataracts, skin cancer
- ▶ UV-B and UV-C absorbed by the ozone layer
- ▶ Ozone depletion increases UV-B surface exposure

**Ozone layer:** absorbs most UV-B and virtually all UV-C, protecting life at Earth's surface.



# Outline of Part III: The Electromagnetic Spectrum

The Spectrum as a Continuum

Radio Waves and Microwaves

Infrared, Visible, and Ultraviolet

X-Rays and Gamma Rays

# X-Rays

## Properties

- ▶ Wavelengths:  $\sim 0.01$  nm to  $\sim 10$  nm
- ▶ Produced by rapid deceleration of high-energy electrons (in an X-ray tube)
- ▶ Highly penetrating through soft tissue; absorbed by bone (high- $Z$  nuclei)
- ▶ Ionizing radiation: can damage DNA

## Applications

- ▶ Medical X-ray imaging and CT scans
- ▶ Dental X-rays
- ▶ X-ray crystallography (determining molecular structure)
- ▶ Airport security scanning

## X-ray vs. gamma ray:

Distinguished by *origin*, not wavelength:

- ▶ X-rays: produced by **electrons** through *bremssstrahlung*:  
German for “braking radiation”.  
bremsen: to brake  
der Strahl: ray (of light)
- ▶ Gamma rays: produced by **nuclear** processes

Overlapping wavelength ranges exist.

# Gamma Rays

## Properties

- ▶  $\lambda < 0.01 \text{ nm}$ ; produced by nuclear decay
- ▶ Most penetrating EM radiation
- ▶ Requires cm of lead or meters of concrete to shield
- ▶ Strongly ionizing — highly damaging to tissue

## Applications

- ▶ Cancer radiation therapy
- ▶ PET scans; sterilization of equipment
- ▶ Astrophysics: gamma-ray bursts, neutron stars

## Penetration Hierarchy

- ▶ **Radio:** no shielding needed
- ▶ **UV:** skin and clothing
- ▶ **X-rays:** lead aprons
- ▶ **Gamma:** cm of lead or m of concrete

# A Bridge Toward Quantum Physics

## Two Levels of Description for EM Waves

- ▶ **Classical:** intensity depends on amplitude squared ( $E_0^2$ )
- ▶ **Quantum:** energy is also carried in discrete packets — **photons**

## Two Independent Quantities

- ▶ **Frequency  $f$ :** sets energy *per photon*
- ▶ **Amplitude  $E_0$ :** sets total intensity
- ▶ Higher  $f \Rightarrow$  each photon can ionize atoms

X-rays and radio waves can have the same intensity, yet X-rays are far more biologically dangerous because each X-ray photon carries enough energy to ionize an atom.

## Key distinction:

*Intensity* = total energy per area per time

*Photon energy* =  $hf$  determines whether a single quantum can break a bond

Both matter for biological effects.

## Check Your Understanding

Which color of visible light has the shortest wavelength?

Come to lecture to see the answer.

## Check Your Understanding

Radiation with wavelength  $\lambda = 0.1 \text{ nm}$  is used to image bone structure. What type of EM radiation is this?

Come to lecture to see the answer.

## Check Your Understanding

Order these from **lowest** to **highest** frequency:

visible light    infrared    gamma rays    microwaves  
ultraviolet    X-rays

Come to lecture to see the answer.

## Check Your Understanding

If the ozone layer were significantly depleted, what would be the primary consequence at Earth's surface?

Come to lecture to see the answer.



## Part IV

# Energy in Electromagnetic Waves

## Outline of Part IV: Energy in Electromagnetic Waves

## Intensity and Field Amplitudes

# Outline of Part IV: Energy in Electromagnetic Waves

Intensity and Field Amplitudes

# Electromagnetic Waves Carry Energy

## Energy Transport by EM Waves

- ▶ EM waves transfer energy from source to receiver without matter transport
- ▶ Energy is stored in the electric and magnetic fields themselves
- ▶ The **direction** of energy flow is  $\vec{E} \times \vec{B}$  (same as propagation direction)
- ▶ Both fields contribute *equally* to the total energy density

## Energy Density in Fields

$$u_E = \frac{1}{2}\epsilon_0 E^2 \quad (\text{electric})$$

$$u_B = \frac{B^2}{2\mu_0} \quad (\text{magnetic})$$

In an EM wave:  $u_E = u_B$  at all times.

The equal partition of energy between  $\vec{E}$  and  $\vec{B}$  is a direct consequence of:

$$c = \frac{E}{B} = \frac{1}{\sqrt{\mu_0\epsilon_0}}$$

# Intensity

Intensity  $I$  — SI Unit:  $\frac{\text{W}}{\text{m}^2}$

Intensity is the average power delivered per unit area perpendicular to the direction of propagation:

$$I = \frac{P}{A}$$

## Variable Definitions

- ▶  $I$ : intensity ( $\frac{\text{W}}{\text{m}^2}$ )
- ▶  $P$ : power (W)
- ▶  $A$ : area through which wave passes ( $\text{m}^2$ )

## Examples of Intensity

- ▶ Sunlight at Earth's surface:  
 $\approx 1000 \frac{\text{W}}{\text{m}^2}$
- ▶ Laser pointer:  $\sim 10^4$ – $10^6 \frac{\text{W}}{\text{m}^2}$
- ▶ Dim starlight:  $\sim 10^{-10} \frac{\text{W}}{\text{m}^2}$

# Average EM Wave Intensity

Intensity in Terms of Field Amplitudes — SI Unit:  $\frac{\text{W}}{\text{m}^2}$

For a sinusoidal EM wave with peak fields  $E_0$  and  $B_0$ :

$$I_{\text{ave}} = \frac{c\varepsilon_0 E_0^2}{2} = \frac{cB_0^2}{2\mu_0} = \frac{E_0 B_0}{2\mu_0}$$

## Physical Interpretation

- ▶ Intensity scales as  $E_0^2$  (amplitude squared)
- ▶ The factor  $\frac{1}{2}$  comes from time-averaging  $\sin^2(\omega t)$
- ▶ Doubling  $E_0$  quadruples intensity

## ☠ Peak vs. Average

Since the instantaneous intensity oscillates between 0 and  $I_{\text{max}}$ :

$$I_{\text{max}} = 2 I_{\text{ave}}$$

Detectors typically measure  $I_{\text{ave}}$ .

# Intensity Follows the Inverse-Square Law over Space

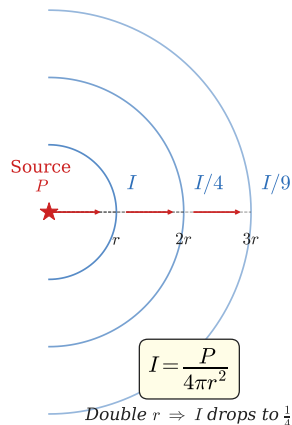
## Point Source in Free Space

A source of total power  $P$  radiates uniformly in all directions. At distance  $r$ , the power is spread over a sphere of area  $A = 4\pi r^2$ :

$$I = \frac{P}{4\pi r^2}$$

## Consequence

- ▶ Double the distance  $\Rightarrow$  intensity falls to  $\frac{1}{4}$
- ▶ Triple the distance  $\Rightarrow$  intensity falls to  $\frac{1}{9}$
- ▶ The *total* power  $P$  is conserved — it just spreads over a larger area



## Example: Intensity and Field Amplitudes from a Lamp

### Fields from a Lamp

100 W bulb, uniform radiation,  $r = 1.00$  m: find  $I$ ,  $E_0$ ,  $B_0$ .

$$I = \frac{P}{4\pi r^2} = \frac{100}{4\pi(1)^2}$$

$$I \approx 7.96 \frac{\text{W}}{\text{m}^2}$$

$$E_0 = \sqrt{\frac{2I}{c\epsilon_0}}$$

$$= \sqrt{\frac{2(7.96)}{(3 \times 10^8)(8.85 \times 10^{-12})}}$$

$$E_0 \approx 77.5 \frac{\text{V}}{\text{m}}$$

$$B_0 = \frac{E_0}{c} \approx 2.58 \times 10^{-7} \text{ T}$$

$E_0 \approx 78 \frac{\text{V}}{\text{m}}$ ,  $B_0 \approx 260 \text{ nT}$  at 1 m.

Intensity  $\propto E_0^2$ ; fields fall as  $1/r$ .

At  $r = 2$  m: intensity =  $I/4$ .



## Example: Intensity from the Sun

### Solar Radiation at Earth

Sun power  $P = 3.85 \times 10^{26} \text{ W}$ , Earth distance  $r = 1.50 \times 10^{11} \text{ m}$ : find  $I$  and  $E_0$ .

$$I = \frac{P}{4\pi r^2}$$

$$= \frac{3.85 \times 10^{26}}{4\pi(1.50 \times 10^{11})^2}$$

$$I \approx 1360 \frac{\text{W}}{\text{m}^2}$$

$$E_0 = \sqrt{\frac{2I}{c\epsilon_0}}$$

$$= \sqrt{\frac{2(1360)}{(3.00 \times 10^8)(8.85 \times 10^{-12})}}$$

$$E_0 \approx 1012 \frac{\text{V}}{\text{m}}$$

Solar constant  $\approx 1360 \frac{\text{W}}{\text{m}^2}$  — basis for solar panel design.

Atmosphere absorbs  $\sim 30\%$ , leaving  $\sim 1000 \frac{\text{W}}{\text{m}^2}$  at Earth's surface.

## Example: Using the Intensity Formula

### Electric Field from Intensity

A microwave oven produces an average intensity of  $I_{\text{ave}} = 5000 \frac{\text{W}}{\text{m}^2}$  inside the cooking cavity. Find the peak electric field  $E_0$  and peak magnetic field  $B_0$ .

$$E_0 = \sqrt{\frac{2 I_{\text{ave}}}{c \epsilon_0}} \quad \because I_{\text{ave}} = \frac{c \epsilon_0 E_0^2}{2}$$

$$= \sqrt{\frac{2(5000)}{(3.00 \times 10^8)(8.85 \times 10^{-12})}}$$

$$E_0 \approx 1940 \frac{\text{V}}{\text{m}}$$

$$B_0 = \frac{E_0}{c} = \frac{1940}{3.00 \times 10^8}$$

$$B_0 \approx 6.47 \mu\text{T}$$

Nearly  $2000 \frac{\text{V}}{\text{m}}$  peak — that's why metal in a microwave sparks.

$$I_{\text{max}} = 2 I_{\text{ave}} = 10\,000 \frac{\text{W}}{\text{m}^2}.$$

Note:  $E_0 \propto \sqrt{I}$ , not  $I$ .

## Check Your Understanding

A point source of light has power  $P$ . If you move from a distance  $r$  to a distance  $3r$ , by what factor does the intensity change?

Come to lecture to see the answer.

## Check Your Understanding

The peak electric field  $E_0$  of an EM wave is doubled. By what factor does the average intensity change?

Come to lecture to see the answer.

## Check Your Understanding

At equal intensity, why are X-rays more biologically damaging than radio waves?

Come to lecture to see the answer.

## Part V

# Chapter Synthesis

# Outline of Part V: Chapter Synthesis

The Unifying Theme

# Outline of Part V: Chapter Synthesis

The Unifying Theme



## The Unifying Theme of Chapter 24

Changing  $\vec{E} \Leftrightarrow$  Changing  $\vec{B} \Rightarrow$  self-propagating EM wave  
 $\Rightarrow$  light, radio, X-rays, and all EM radiation are one phenomenon

### The Four Pillars of Chapter 24

- ▶ **Prediction:** Maxwell predicted waves at speed  $c$
- ▶ **Production:** accelerating charges radiate
- ▶ **Spectrum:** one continuum, organized by  $f$
- ▶ **Energy:**  $I \propto E_0^2$ , falls as  $1/r^2$

### Historical Legacy

- ▶ 1865: Maxwell predicts EM waves
- ▶ 1888: Hertz detects radio waves experimentally
- ▶ Opens the modern era of wireless communication and quantum mechanics

# Key Equations of Chapter 24

## Wave Properties

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = f\lambda = \frac{E}{B}$$

## Intensity

$$I = \frac{P}{4\pi r^2}$$

$$I_{\text{ave}} = \frac{c\epsilon_0 E_0^2}{2}$$

$$I_{\text{max}} = 2 I_{\text{ave}}$$

## Spectrum (low $\rightarrow$ high $f$ )

- ▶ Radio, Microwave, Infrared
- ▶ Visible: R $\rightarrow$ V = low $\rightarrow$ high  $f$
- ▶ UV, X-ray, Gamma ray

## Constants:

$$c = 3.00 \times 10^8 \frac{\text{m}}{\text{s}}$$

$$\mu_0 = 4\pi \times 10^{-7} \frac{\text{T}\cdot\text{m}}{\text{A}}$$

$$\epsilon_0 = 8.85 \times 10^{-12} \frac{\text{C}^2}{\text{N}\cdot\text{m}^2}$$

# Looking Ahead

## What Comes Next

- ▶ **Geometric Optics** (Ch. 25–26): reflection, refraction, lenses and mirrors
- ▶ **Wave Optics** (Ch. 27): interference and diffraction
- ▶ **Quantum Mechanics** (Ch. 29–31): EM waves carry energy in photons;  
 $E_{\text{photon}} = hf$  (Note: this class only covers up to Ch. 28)

## The Big Picture

Chapter 24 is the *bridge* between classical and modern physics.

Maxwell's equations are among the most fundamental and enduring laws in all of physics.

## One phenomenon, many manifestations:

Radio, microwave, IR, visible, UV, X-rays, gamma rays — all are oscillating  $\vec{E}$  and  $\vec{B}$  fields, differing only in frequency.